

Derivative-Map Dynamics of Finite- N Central-Limit Shape Collapse: Stationary–Inflection Skeletons, Characteristic Functions, and Finite-Window Mode-Death Thresholds

David Hongkai Shen

M.S. candidate in Mathematics in Finance, Courant Institute of Mathematical Sciences,
New York University, New York, NY, USA
hs4475@nyu.edu

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Abstract

This paper studies the finite- N derivative geometry of standardized sums in the central-limit regime. The analysis is not intended to provide a new central limit theorem; rather, it describes how the density of the standardized sum

$$Z_N = (X_1 + \cdots + X_N - N\mu)/(\sigma\sqrt{N})$$

changes its stationary and curvature structure as N increases. Here $X_1, \dots, X_N$ are independent copies of X , with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \text{Var}(X) > 0$. When Z_N has a sufficiently smooth density f_N , the zero sets and signs of f'_N and f''_N determine local modes, valleys, and inflection points. These objects define a stationary–inflection skeleton for the finite- N density. The standard Gaussian endpoint has one stationary point, two inflection points, and the four-zone sign word

$$(+, +) \rightarrow (+, -) \rightarrow (-, -) \rightarrow (-, +),$$

written as $I \rightarrow IV \rightarrow III \rightarrow II$. Characteristic functions provide the analytic mechanism: convolution becomes multiplication, standardization suppresses cumulants of order at least three, and Fourier inversion transfers this decay to density derivatives under suitable integrability assumptions. The paper formulates derivative local-CLT inputs, establishes deterministic skeleton-stability and zero-motion statements on finite windows, derives sufficient mode-death thresholds excluding off-center stationary points, records Edgeworth-based zero-location formulas, and describes smoothing extensions for discrete laws. The resulting representation may be used as a finite-window diagnostic for multimodal-to-Gaussian shape convergence, with possible applications in statistical learning, numerical probability, signal aggregation, and financial risk aggregation. A lightweight companion Python implementation illustrates the diagnostic pipeline and generates the figures and metric tables used in the numerical section.

Keywords: central limit theorem; local central limit theorem; derivative maps; convolution powers; multimodal densities; characteristic functions; Edgeworth expansion; stationary points; inflection points; density-shape diagnostics; numerical probability.

JEL classification: C02; C15; C46; C63.

1 Introduction

Let $X_1, X_2, \dots$ be independent copies of a real-valued random variable X , with mean $\mu = \mathbb{E}[X]$ and variance $\sigma^2 = \text{Var}(X) > 0$. The classical central limit theorem states that [3, 5, 7]

$$Z_N := \frac{X_1 + \dots + X_N - N\mu}{\sigma\sqrt{N}} \xrightarrow{d} G \sim N(0, 1),$$

where $\xrightarrow{d}$ denotes convergence in distribution and G is a standard Gaussian random variable. This convergence statement alone does not describe the finite- N derivative geometry of the density. In particular, when the initial law is mixture-like or multimodal, the density of Z_N may contain several local maxima, intervening local minima, and additional inflection points before it approaches the one-mode Gaussian shape.

The present paper, *Derivative-Map Dynamics of Finite- N Central-Limit Shape Collapse* (DMD-CLT), examines the finite- N derivative geometry of central-limit shape collapse. Whenever Z_N has a sufficiently smooth density f_N , define

$$\mathcal{S}_N := \{x \in \mathbb{R} : f'_N(x) = 0\}, \quad \mathcal{I}_N := \{x \in \mathbb{R} : f''_N(x) = 0\}.$$

The set $\mathcal{S}_N$ records stationary points, including local maxima and local minima. The set $\mathcal{I}_N$ records inflection points. The pair $(\mathcal{S}_N, \mathcal{I}_N)$ is the stationary–inflection skeleton of the finite- N density. Here and below, $\mathbb{R}$ denotes the real line.

The analysis treats central-limit convergence as a sequence of derivative-level shape transitions. Standardized convolution is generated on the characteristic-function side [1, 5, 6], while the density-side effect is measured through the zero sets and signs of f'_N and f''_N . The Gaussian endpoint has a particularly simple skeleton: one stationary point at 0, inflection points at ± 1 , and the derivative sign word

$$I \rightarrow IV \rightarrow III \rightarrow II.$$

A multimodal convolution density may have a longer stationary–inflection word at small or moderate N . The analysis focuses on conditions under which derivative convergence is sufficient to reduce this word to the Gaussian word on controlled windows.

The argument separates probabilistic convergence from deterministic shape stability. The probabilistic component supplies derivative-level convergence estimates, such as derivative local limit or Edgeworth-type approximations. Once these estimates are available, derivative-map stabilization can be derived from deterministic principles: zero-motion bounds, exclusion thresholds away from the Gaussian skeleton, and finite-window mode-death criteria. Thus the role of the probabilistic limit theorem in this formulation is to provide quantitative error bounds, while the disappearance of extra modes is governed by deterministic derivative-map inequalities. In this sense, the contribution is best interpreted as a derivative-map representation of finite- N central-limit shape convergence rather than a modification of the central limit theorem itself.

The numerical component is deliberately modest. It is not used as a substitute for the analytic results; it is included to show that the stationary–inflection objects are directly computable from characteristic functions. The companion script reconstructs f_N , f'_N , and f''_N by exact finite Gaussian-mixture convolution for the built-in examples, with a direct Fourier-quadrature route retained as an optional diagnostic check. It then detects stationary and inflection zeros, records sign words, and exports tolerance-dependent diagnostic metrics.

Section 10 is the main deterministic component of the paper. It converts derivative-level error estimates into explicit finite-window consequences, including controlled displacement of simple Gaussian skeleton zeros, exclusion of off-center stationary points, and sufficient conditions for collapse to a single central mode. The preceding Fourier and local-limit sections explain how the derivative errors arise and decay; Section 10 explains why errors below quantitative thresholds force the finite-window mode-death transition.

2 Geometric representation and Fourier mechanism

The derivative-map representation has two components. The first component is geometric: Figures 1–3 summarize the four sign regions of (f', f'') and the stationary–inflection skeleton of the standard normal density. The second component is analytic: characteristic functions explain why standardized convolution removes higher-cumulant contributions and produces the Gaussian limiting skeleton.

1. Figure 1 records the four possible local signs of (f', f'') :

$$(+, +), \quad (+, -), \quad (-, -), \quad (-, +).$$

These correspond respectively to increasing–convex, increasing–concave, decreasing–concave, and decreasing–convex regions.

2. Figure 2 plots the standard normal density together with its first two derivatives. This display indicates that the Gaussian density has a unique stationary point at $x = 0$ and two inflection points at $x = \pm 1$.
3. Figure 3 places the same transition points directly on the Gaussian density. From left to right, the Gaussian derivative sign word is

$$I \rightarrow IV \rightarrow III \rightarrow II.$$

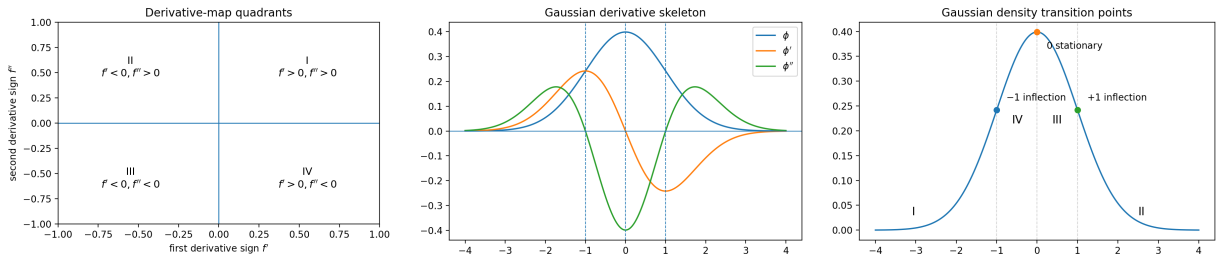


Figure 1–3: Derivative-map quadrants, the Gaussian derivative skeleton, and the Gaussian density transition points. The three panels are arranged together because they describe the same stationary–inflection structure from complementary views. The digital version may be enlarged for detailed inspection.

2.1 Characteristic-function mechanism

The analytic component is the Fourier-side derivation. In the Fourier formulas, i denotes the imaginary unit, $\mathbb{E}$ denotes expectation, and $\mathbb{P}$ denotes probability. For an absolutely continuous

random variable with density f_X , the characteristic function is

$$\varphi_X(t) = \int_{-\infty}^{\infty} e^{itx} f_X(x) dx. \quad (1)$$

For a discrete random variable,

$$\varphi_X(t) = \sum_x e^{itx} \mathbb{P}(X = x). \quad (2)$$

For a Gaussian random variable $X \sim N(\mu, \sigma^2)$,

$$\varphi_X(t) = \exp\left(i\mu t - \frac{1}{2}\sigma^2 t^2\right). \quad (3)$$

Let

$$Y = \frac{X - \mu}{\sigma}, \quad Z_N = \frac{1}{\sqrt{N}} \sum_{k=1}^N Y_k,$$

where $Y_1, \dots, Y_N$ are iid copies of Y . On the Fourier side, convolution of iid sums becomes multiplication of characteristic functions:

$$\varphi_{Z_N}(t) = \left[\varphi_Y\left(\frac{t}{\sqrt{N}}\right) \right]^N. \quad (4)$$

Assuming the relevant cumulants exist, the logarithmic cumulant expansion gives

$$\log \varphi_{Z_N}(t) = -\frac{t^2}{2} + \sum_{r \geq 3} \kappa_r(Y) \frac{(it)^r}{r!} N^{1-r/2}. \quad (5)$$

The terms with $r \geq 3$ vanish as $N \rightarrow \infty$. Thus the non-Gaussian part of the Fourier exponent disappears, leaving the quadratic Gaussian exponent:

$$\log \varphi_{Z_N}(t) \longrightarrow -\frac{t^2}{2}, \quad \varphi_{Z_N}(t) \longrightarrow e^{-t^2/2}. \quad (6)$$

For an unstandardized Gaussian target, the corresponding exponent is

$$i\mu t - \frac{1}{2}\sigma^2 t^2.$$

The Fourier-side mechanism can therefore be summarized as

$$\begin{aligned} &\text{convolution of iid sums} \longrightarrow \text{multiplication of characteristic functions} \\ &\longrightarrow \text{higher-cumulant decay under } \sqrt{N}\text{-normalization} \longrightarrow \text{Gaussian characteristic function.} \end{aligned} \quad (7)$$

Through Fourier inversion, this becomes the density-side implication

$$\begin{aligned} &\text{Fourier side: convolution and cumulant decay} \\ &\implies \text{density side: convergence of derivative maps.} \end{aligned} \quad (8)$$

This bridge explains why the derivative-map skeleton is an appropriate object for studying finite- N central-limit behavior.

3 Fourier convention and standardized convolution

Throughout the paper the characteristic function of a random variable Y is

$$\varphi_Y(t) := \mathbb{E}e^{itY}, \quad t \in \mathbb{R}.$$

If Y has density f_Y , the inversion convention is

$$f_Y(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \varphi_Y(t) dt,$$

whenever the integral is justified. Here dt denotes integration with respect to the Fourier variable t , and $L^1(\mathbb{R})$ below denotes the class of integrable functions on the real line.

Let

$$Y := \frac{X - \mu}{\sigma}, \quad \mathbb{E}Y = 0, \quad \text{Var}(Y) = 1.$$

Then

$$Z_N = \frac{Y_1 + \cdots + Y_N}{\sqrt{N}}.$$

If $\psi(t) := \varphi_Y(t)$, then

$$\varphi_{Z_N}(t) = \left[\psi\left(\frac{t}{\sqrt{N}}\right) \right]^N. \quad (9)$$

In terms of the original variable X ,

$$\varphi_{Z_N}(t) = \left[\varphi_X\left(\frac{t}{\sigma\sqrt{N}}\right) \exp\left(-i\frac{\mu t}{\sigma\sqrt{N}}\right) \right]^N. \quad (10)$$

Equation (10) is the basic Fourier representation. Direct N -fold convolution in x -space is replaced by exponentiation in t -space.

If $\varphi_{Z_N} \in L^1(\mathbb{R})$ and $|t|^k |\varphi_{Z_N}(t)| \in L^1(\mathbb{R})$, then the k th derivative of f_N can be written as

$$f_N^{(k)}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} (-it)^k e^{-itx} \varphi_{Z_N}(t) dt. \quad (11)$$

Thus derivatives in x -space are generated by the Fourier multiplier $(-it)^k$. The formula is used only in regimes where the stated integrability assumptions justify differentiating under the Fourier integral. In particular, if $|t|^k |\varphi_{Z_N}(t)| \in L^1(\mathbb{R})$, then the right-hand side of (11) defines a bounded continuous version of $f_N^{(k)}$.

4 The Gaussian derivative skeleton

Let

$$\phi(x) := \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

be the standard normal density. Its first three derivatives are

$$\phi'(x) = -x\phi(x), \quad (12)$$

$$\phi''(x) = (x^2 - 1)\phi(x), \quad (13)$$

$$\phi'''(x) = -(x^3 - 3x)\phi(x). \quad (14)$$

Here and below, $\mathcal{Z}(g) := \{x \in \mathbb{R} : g(x) = 0\}$ denotes the zero set of a function g . Therefore

$$\mathcal{Z}(\phi') = \{0\}, \quad \mathcal{Z}(\phi'') = \{-1, 1\}, \quad \mathcal{Z}(\phi''') = \{-\sqrt{3}, 0, \sqrt{3}\}.$$

The analysis uses ϕ' and ϕ'' as the primary skeleton maps. The third derivative ϕ''' may be added when curvature-acceleration information is needed.

The signs of (ϕ', ϕ'') on the four Gaussian intervals are as follows:

Interval	ϕ'	ϕ''	label
$(-\infty, -1)$	+	+	<i>I</i>
$(-1, 0)$	+	-	<i>IV</i>
$(0, 1)$	-	-	<i>III</i>
$(1, \infty)$	-	+	<i>II</i>

Hence the canonical Gaussian derivative sign word is

$$I \rightarrow IV \rightarrow III \rightarrow II. \tag{15}$$

This word is used as the limiting skeleton. A multimodal density may have a longer word because additional local maxima, local minima, and inflection points create additional sign intervals. Finite- N convergence toward the Gaussian skeleton is represented as reduction of this longer word to (15) on appropriate windows.

5 Derivative sign maps and sign words

Definition 5.1 (Derivative quadrant map). Let f be twice continuously differentiable on an interval $J \subseteq \mathbb{R}$. Here sgn denotes the usual sign function. At points where $f'(x)f''(x) \neq 0$, define

$$Q_f(x) := Q(\text{sgn } f'(x), \text{sgn } f''(x)),$$

where

$$Q(+, +) = I, \quad Q(+, -) = IV, \quad Q(-, -) = III, \quad Q(-, +) = II.$$

Definition 5.2 (Stationary-inflection sign word). Let B_f be the ordered set of distinct points in J where f' or f'' vanishes. If this set is finite, then $J \setminus B_f$ is a union of open intervals. The sign word $W_J(f)$ is obtained by moving from left to right across those intervals and recording Q_f on each interval. Since f' and f'' are continuous and nonzero on each component of $J \setminus B_f$, their signs are constant on that component. In numerical implementations, nearly multiple zeros may be merged by a specified tolerance, but the mathematical definition uses the distinct zero set.

For the standard Gaussian over $J = \mathbb{R}$, the word is exactly

$$W_{\mathbb{R}}(\phi) = I \rightarrow IV \rightarrow III \rightarrow II.$$

For a density with two separated modes, the observed word may be a longer word with repeated peak-valley structure. The analysis concerns the dependence of $W_J(f_N)$ on N .

Remark 5.1 (Role of the first two derivatives). The pair (f', f'') records monotonicity and curvature. A zero of f' gives a stationary point, while a zero of f'' gives an inflection point. These objects are sufficient to distinguish additional modes from the one-mode Gaussian skeleton. Higher derivatives may refine the analysis, but the finite- N derivative sign map is already defined by (f', f'') .

6 Cumulant collapse on the characteristic-function side

Assume $Y = (X - \mu)/\sigma$ has cumulants $\kappa_m(Y)$ for the required orders; the characteristic-function and cumulant notation follows the classical convention in [1, 5]. The following expansion is understood for fixed t after choosing a continuous branch of $\log \psi$ in a neighborhood of the origin where the Taylor expansion is valid. Since $\kappa_1(Y) = 0$ and $\kappa_2(Y) = 1$,

$$\log \psi(u) = -\frac{u^2}{2} + \sum_{m=3}^r \frac{\kappa_m(Y)}{m!} (iu)^m + o(|u|^r), \quad u \rightarrow 0. \quad (16)$$

Substituting $u = t/\sqrt{N}$ and multiplying by N gives, for fixed t and with the usual locally uniform interpretation when the Taylor remainder is locally uniform,

$$\begin{aligned} \log \varphi_{Z_N}(t) &= N \log \psi \left(\frac{t}{\sqrt{N}} \right) \\ &= -\frac{t^2}{2} + \sum_{m=3}^r \frac{\kappa_m(Y)}{m!} N^{1-m/2} (it)^m + o \left(N^{1-r/2} |t|^r \right). \end{aligned} \quad (17)$$

Thus the m th cumulant of Z_N is

$$\kappa_m(Z_N) = \kappa_m(Y) N^{1-m/2} = \frac{\kappa_m(X)}{\sigma^m N^{m/2-1}}, \quad m \geq 3. \quad (18)$$

Equation (18) gives the cumulant-scaling mechanism behind finite- N convergence of the derivative skeleton. Skewness decays at order $N^{-1/2}$, the fourth cumulant decays at order N^{-1} , and higher cumulants decay even faster. The only non-vanishing limiting cumulant is the second cumulant, so

$$\varphi_{Z_N}(t) \rightarrow e^{-t^2/2}.$$

This characteristic-function limit serves as the analytic source of the Gaussian derivative skeleton.

7 Edgeworth form and derivative corrections

Classical Edgeworth expansions provide the polynomial corrections used below [1, 2, 7, 8, 9]. Let He_m denote the probabilists' Hermite polynomial, defined by

$$\text{He}_m(x) = (-1)^m \frac{\phi^{(m)}(x)}{\phi(x)}.$$

The first few are

$$\begin{aligned} \text{He}_0 &= 1, & \text{He}_1 &= x, & \text{He}_2 &= x^2 - 1, & \text{He}_3 &= x^3 - 3x, \\ \text{He}_4 &= x^4 - 6x^2 + 3, & \text{He}_5 &= x^5 - 10x^3 + 15x, & \text{He}_6 &= x^6 - 15x^4 + 45x^2 - 15. \end{aligned}$$

Under the usual moment, smoothness, and non-lattice hypotheses required for a derivative Edgeworth expansion, and when the expansion and its required derivatives are valid up to order N^{-1} uniformly on the window under consideration, the density has the form

$$f_N(x) = \phi(x) + N^{-1/2}A(x) + N^{-1}B(x) + o(N^{-1}), \quad (19)$$

where

$$A(x) = \frac{\lambda_3}{6} \text{He}_3(x) \phi(x), \quad (20)$$

$$B(x) = \left(\frac{\lambda_4}{24} \text{He}_4(x) + \frac{\lambda_3^2}{72} \text{He}_6(x) \right) \phi(x). \quad (21)$$

Here

$$\lambda_3 := \kappa_3(Y), \quad \lambda_4 := \kappa_4(Y)$$

are the standardized third and fourth cumulants. Differentiating (19) gives

$$f_N^{(k)}(x) = \phi^{(k)}(x) + N^{-1/2} A^{(k)}(x) + N^{-1} B^{(k)}(x) + o(N^{-1}) \quad (22)$$

for $k = 0, 1, 2$ under derivative Edgeworth conditions. This equation connects cumulant decay with derivative-map convergence. The $o(N^{-1})$ term in (22) is interpreted uniformly on the fixed finite window used for the zero-location or sign-stability argument.

Remark 7.1 (Sign convention). The sign in (19) follows the Fourier convention

$$f(x) = \frac{1}{2\pi} \int e^{-itx} \varphi(t) dt.$$

Under this convention, the inverse transform of $(it)^m e^{-t^2/2}$ is $\text{He}_m(x) \phi(x)$.

8 Derivative local-CLT input

A fully general theorem asserting that every multimodal initial density eventually has exactly the Gaussian global sign word is not valid without additional regularity and tail assumptions. The mathematical argument is therefore separated into two parts:

1. a derivative local-CLT input, which gives convergence of f_N , f'_N , and f''_N to the corresponding Gaussian derivatives on controlled regions;
2. a skeleton-stability theorem, which converts derivative convergence into stability of stationary points, inflection points, and sign words away from transition boundaries.

Assumption 8.1 (Derivative local-CLT condition). For some integer $K \geq 3$, the standardized convolution density f_N exists for all sufficiently large N , is K times continuously differentiable, and satisfies

$$\epsilon_{N,K}(R) := \max_{0 \leq k \leq K} \sup_{|x| \leq R} |f_N^{(k)}(x) - \phi^{(k)}(x)| \rightarrow 0 \quad (23)$$

for every finite $R > 0$. The quantity $\epsilon_{N,K}(R)$ is the maximum finite-window derivative error up to order K .

This condition follows from standard derivative versions of local central limit and Edgeworth theory under common smoothness, moment, and non-lattice assumptions [7, 8, 6]. A standard sufficient condition is that Y has sufficiently many finite moments, its characteristic function satisfies a Cramer-type non-lattice condition, and for some sufficiently large convolution power the required Fourier integrals with multipliers $|t|^k$ are integrable. Smooth finite Gaussian mixtures satisfy such conditions.

Proposition 8.1 (Fourier route to derivative convergence). Fix an integer $K \geq 0$. Suppose that Z_N has a density f_N and that, for each $0 \leq k \leq K$, Fourier inversion may be differentiated k times as in (11). Assume further that

$$\varphi_{Z_N}(t) \rightarrow e^{-t^2/2} \quad \text{for every } t \in \mathbb{R},$$

and that for each $0 \leq k \leq K$ there exists an integrable function $G_k \in L^1(\mathbb{R})$ such that, for all sufficiently large N ,

$$|t|^k \left| \varphi_{Z_N}(t) - e^{-t^2/2} \right| \leq G_k(t) \quad (t \in \mathbb{R}).$$

Then

$$\sup_{x \in \mathbb{R}} |f_N^{(k)}(x) - \phi^{(k)}(x)| \rightarrow 0, \quad 0 \leq k \leq K.$$

Proof. By Fourier inversion,

$$f_N^{(k)}(x) - \phi^{(k)}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} (-it)^k e^{-itx} \left(\varphi_{Z_N}(t) - e^{-t^2/2} \right) dt.$$

Taking absolute values removes the oscillatory factor and gives the uniform estimate

$$\sup_x |f_N^{(k)}(x) - \phi^{(k)}(x)| \leq \frac{1}{2\pi} \int_{\mathbb{R}} |t|^k \left| \varphi_{Z_N}(t) - e^{-t^2/2} \right| dt.$$

The integrand converges pointwise to zero and is dominated by $G_k \in L^1(\mathbb{R})$. Dominated convergence therefore implies that the right-hand side tends to zero. This proves the claim.

9 Stability of the Gaussian stationary–inflection skeleton

The following theorem gives a central deterministic component of the derivative-map construction. It states that once f'_N and f''_N are close to ϕ' and ϕ'' on a central window, the Gaussian sign structure is forced on that window away from small neighborhoods of $-1, 0, 1$, and the main stationary and inflection points are uniquely located near the Gaussian ones.

Theorem 9.1 (Central-window Gaussian skeleton stability). Assume Assumption 8.1 with $K \geq 3$. Fix $R > 2$ and choose $\eta \in (0, 1/4)$. For all sufficiently large N , the following statements hold.

1. There is exactly one zero m_N of f'_N in $(-\eta, \eta)$. Moreover, $m_N \rightarrow 0$.
2. There is exactly one zero $\iota_{N,-}$ of f''_N in $(-1 - \eta, -1 + \eta)$ and exactly one zero $\iota_{N,+}$ of f''_N in $(1 - \eta, 1 + \eta)$. Moreover,

$$\iota_{N,-} \rightarrow -1, \quad \iota_{N,+} \rightarrow 1.$$

3. On the compact set

$$K_{R,\eta} := [-R, R] \setminus [(-1 - \eta, -1 + \eta) \cup (-\eta, \eta) \cup (1 - \eta, 1 + \eta)],$$

the signs of (f'_N, f''_N) equal the signs of (ϕ', ϕ'') for all sufficiently large N .

4. No wrong-order zero is introduced inside the three transition neighborhoods: for all sufficiently large N , f'_N has no zero in the two inflection neighborhoods $(-1 - \eta, -1 + \eta)$ and $(1 - \eta, 1 + \eta)$, while f''_N has no zero in the central mode neighborhood $(-\eta, \eta)$.

Proof. For the stationary point, choose $\eta < 1$. On $[-\eta, \eta]$, the Gaussian second derivative satisfies

$$\phi''(x) = (x^2 - 1)\phi(x) < 0.$$

Therefore there is $c_0 > 0$ such that $\phi''(x) \leq -c_0$ on this interval. By derivative convergence, $f_N''(x) \leq -c_0/2$ there for all large N , so f_N' is strictly decreasing on $[-\eta, \eta]$. Also $\phi'(-\eta) > 0$ and $\phi'(\eta) < 0$, and convergence of f_N' implies $f_N'(-\eta) > 0$ and $f_N'(\eta) < 0$ for large N . Hence f_N' has exactly one zero $m_N \in (-\eta, \eta)$. To prove $m_N \rightarrow 0$, fix any $\varepsilon > 0$ and repeat the same argument on $[-\varepsilon, \varepsilon]$ after replacing ε by a smaller number if necessary so that it is less than 1. For all sufficiently large N , there is a unique zero of f_N' in that smaller interval. By uniqueness in the original central interval, this zero must be m_N . Hence $|m_N| < \varepsilon$ for all sufficiently large N .

For the inflection point near 1, note that

$$\phi'''(1) = 2\phi(1) > 0.$$

Thus, after shrinking η if needed, $\phi'''(x) > c_1 > 0$ on $(1 - \eta, 1 + \eta)$. For large N , $f_N'''(x) > c_1/2$ on this interval, so f_N'' is strictly increasing there. Moreover, $\phi''(1 - \eta) < 0$ and $\phi''(1 + \eta) > 0$, hence the same signs hold for f_N'' for all sufficiently large N . Therefore f_N'' has exactly one zero in $(1 - \eta, 1 + \eta)$. Its convergence to 1 follows by repeating the same argument on arbitrarily small neighborhoods of 1 and using uniqueness in the original neighborhood. The proof near -1 is analogous: since $\phi'''(-1) = -2\phi(1) < 0$ and $\eta < 1/4 < \sqrt{3} - 1$, ϕ''' remains strictly negative on $(-1 - \eta, -1 + \eta)$; hence f_N'' is strictly decreasing there for large N , and the endpoint signs force a unique zero. The same shrinking-neighborhood argument gives convergence of this zero to -1 .

The wrong-order zero exclusion follows from the fact that ϕ' is bounded away from zero on the two inflection neighborhoods and ϕ'' is bounded away from zero on the central mode neighborhood. Uniform convergence of the corresponding derivatives then prevents f_N' from vanishing near ± 1 and prevents f_N'' from vanishing near 0, for all sufficiently large N .

Finally, on $K_{R,\eta}$ neither ϕ' nor ϕ'' vanishes. Since $K_{R,\eta}$ is compact, there is $c_2 > 0$ such that

$$\min_{x \in K_{R,\eta}} \min\{|\phi'(x)|, |\phi''(x)|\} \geq c_2.$$

For large N , $|f_N' - \phi'| < c_2/2$ and $|f_N'' - \phi''| < c_2/2$ on $K_{R,\eta}$, so the signs agree. This proves the theorem.

Remark 9.1 (Central-window versus global sign word). Theorem 9.1 is intentionally stated on $[-R, R]$. Uniform convergence of derivatives on compact sets does not automatically rule out tiny oscillations in far tails where Gaussian derivatives are extremely small. A global statement over all of $\mathbb{R}$ requires additional tail regularity. In computation, this motivates using a probability-mass window or density-threshold window.

Assumption 9.1 (Additional global tail regularity). For some $R > 2$ and all sufficiently large N ,

$$f_N'(x) > 0, \quad f_N''(x) > 0 \quad \text{for } x < -R,$$

and

$$f_N'(x) < 0, \quad f_N''(x) > 0 \quad \text{for } x > R.$$

Corollary 9.1 (Global Gaussian sign word under tail regularity). Under Assumption 8.1 and Assumption 9.1, the global sign word of f_N is

$$I \rightarrow IV \rightarrow III \rightarrow II$$

for all sufficiently large N .

Proof. Choose the central window in Theorem 9.1 with the same R as in Assumption 9.1. The theorem gives exactly one stationary zero near 0, exactly two inflection zeros near ± 1 , agreement of the signs of (f'_N, f''_N) with (ϕ', ϕ'') away from these transition neighborhoods inside $[-R, R]$, and exclusion of wrong-order zeros inside the transition neighborhoods. The tail assumption fixes the signs outside $[-R, R]$ as the Gaussian tail signs. Hence no additional sign intervals occur, and the ordered sign word is the Gaussian word.

10 Rigorous zero-motion and mode-death thresholds

The preceding stability theorem gives an eventual qualitative conclusion: after derivative local-CLT convergence is sufficiently strong on a fixed central window, the Gaussian stationary-inflection skeleton is forced. This section gives quantitative deterministic thresholds converting derivative-map error into zero-motion and elimination of off-center stationary points. The argument is separated from the probabilistic local-CLT input: once an analytic estimate supplies a bound on $f'_N - \phi'$ and $f''_N - \phi''$, the following theorem turns that estimate into a finite- N threshold.

10.1 Zero motion for a simple skeleton zero

The next theorem is a perturbation statement for a simple zero. It applies to stationary points, inflection points, and higher derivative zeros. Here $C^q(I)$ denotes the class of q times continuously differentiable functions on an interval I .

Theorem 10.1 (Zero-motion bound for a simple skeleton zero). Let $g, h \in C^1([\zeta - r, \zeta + r])$, where $r > 0$. Assume that

$$g(\zeta) = 0$$

and that the derivative of g is bounded away from zero on the interval:

$$a := \inf_{|x - \zeta| \leq r} |g'(x)| > 0.$$

Let

$$\delta := \sup_{|x - \zeta| \leq r} |h(x) - g(x)|.$$

If $z \in [\zeta - r, \zeta + r]$ satisfies $h(z) = 0$, then

$$|z - \zeta| \leq \frac{\delta}{a}. \tag{24}$$

If, in addition,

$$\sup_{|x - \zeta| \leq r} |h'(x) - g'(x)| < a,$$

then h has at most one zero in $[\zeta - r, \zeta + r]$.

Proof. Since g' is continuous and $|g'| \geq a > 0$ on a connected interval, g' has a constant sign there. For any x in the interval, the mean value theorem gives

$$|g(x) - g(\zeta)| \geq a|x - \zeta|.$$

Because $g(\zeta) = 0$, this becomes

$$|g(x)| \geq a|x - \zeta|.$$

At a zero z of h ,

$$|g(z)| = |g(z) - h(z)| \leq \delta.$$

Therefore

$$a|z - \zeta| \leq |g(z)| \leq \delta,$$

which proves (24). For uniqueness, if $\sup |h' - g'| < a$, then h' has the same sign as g' throughout the interval. Hence h is strictly monotone there and can have at most one zero.

Corollary 10.1 (Gaussian mode and inflection zero-motion). Let $p \in C^3(\mathbb{R})$.

1. Fix $0 < r_0 < 1$ and define

$$a_0 := \min_{|x| \leq r_0} (-\phi''(x)) = (1 - r_0^2)\phi(r_0) > 0.$$

If m is a zero of p' in $[-r_0, r_0]$ and

$$\delta_1 := \sup_{|x| \leq r_0} |p'(x) - \phi'(x)|,$$

then

$$|m| \leq \frac{\delta_1}{a_0}. \quad (25)$$

If also

$$\sup_{|x| \leq r_0} |p''(x) - \phi''(x)| < a_0,$$

then p' has at most one zero in $[-r_0, r_0]$. In particular, if a central stationary zero exists, it is unique.

2. Fix $0 < r_1 < \min\{1, \sqrt{3} - 1\}$ and define

$$a_+ := \min_{|x-1| \leq r_1} |\phi'''(x)| > 0, \quad a_- := \min_{|x+1| \leq r_1} |\phi'''(x)| > 0.$$

If ι_+ is a zero of p'' in $[1 - r_1, 1 + r_1]$ and

$$\delta_2^+ := \sup_{|x-1| \leq r_1} |p''(x) - \phi''(x)|,$$

then

$$|\iota_+ - 1| \leq \frac{\delta_2^+}{a_+}.$$

If, in addition,

$$\sup_{|x-1| \leq r_1} |p'''(x) - \phi'''(x)| < a_+,$$

then p'' has at most one zero in $[1 - r_1, 1 + r_1]$.

If ι_- is a zero of p'' in $[-1 - r_1, -1 + r_1]$ and

$$\delta_2^- := \sup_{|x+1| \leq r_1} |p''(x) - \phi''(x)|,$$

then

$$|\iota_- + 1| \leq \frac{\delta_2^-}{a_-}.$$

If, in addition,

$$\sup_{|x+1| \leq r_1} |p'''(x) - \phi'''(x)| < a_-,$$

then p'' has at most one zero in $[-1 - r_1, -1 + r_1]$.

Proof. Apply Theorem 10.1 with $g = \phi'$ and $h = p'$ for the mode. Since $g' = \phi'' < 0$ on $[-r_0, r_0]$, the lower derivative bound is a_0 .

For the inflection points, apply Theorem 10.1 with $g = \phi''$ and $h = p''$. The derivative of g is ϕ''' . The restriction $r_1 < \min\{1, \sqrt{3} - 1\}$ prevents the neighborhoods of ± 1 from touching zeros of ϕ''' , so a_+ and a_- are strictly positive.

10.2 Mode-death threshold on a finite window

The zero-motion theorem controls zeros once they are known to be near simple Gaussian skeleton zeros. The next theorem shows that off-center stationary points cannot exist on a fixed finite window after the derivative error falls below an explicit threshold.

Theorem 10.2 (Gaussian mode-death threshold on a finite interval). Fix

$$0 < \eta < 1, \quad R > 1.$$

Define the positive constants

$$\tau_{\eta,R} := \min_{\eta \leq |x| \leq R} |\phi'(x)| > 0, \quad (26)$$

$$\beta_\eta := \min_{|x| \leq \eta} (-\phi''(x)) = (1 - \eta^2)\phi(\eta) > 0. \quad (27)$$

Let $p \in C^2([-R, R])$. Suppose

$$\epsilon_1(R) := \sup_{|x| \leq R} |p'(x) - \phi'(x)| < \tau_{\eta,R}, \quad (28)$$

$$\epsilon_2(\eta) := \sup_{|x| \leq \eta} |p''(x) - \phi''(x)| < \beta_\eta. \quad (29)$$

Then p' has no zero in the off-center compact region

$$\{x : \eta \leq |x| \leq R\}.$$

Moreover, p' has exactly one zero in $(-\eta, \eta)$, and at that zero $p'' < 0$. Thus, within $[-R, R]$, all off-center stationary points are eliminated and the only surviving stationary point is a strict local maximum.

Proof. First consider the off-center region $\eta \leq |x| \leq R$. By definition of $\tau_{\eta,R}$,

$$|\phi'(x)| \geq \tau_{\eta,R}.$$

Since $|p'(x) - \phi'(x)| < \tau_{\eta,R}$, $p'(x)$ must have the same sign as $\phi'(x)$. Hence $p'(x) \neq 0$ throughout this off-center region.

Now consider the central interval $[-\eta, \eta]$. By definition of β_η ,

$$\phi''(x) \leq -\beta_\eta \quad (|x| \leq \eta).$$

Using (29),

$$p''(x) \leq \phi''(x) + |p''(x) - \phi''(x)| < -\beta_\eta + \beta_\eta = 0.$$

Therefore p' is strictly decreasing on $[-\eta, \eta]$.

At $x = -\eta$, $\phi'(-\eta) > 0$ and $|\phi'(-\eta)| \geq \tau_{\eta,R}$. The bound (28) gives $p'(-\eta) > 0$. Similarly $\phi'(\eta) < 0$ and (28) gives $p'(\eta) < 0$. By the intermediate value theorem, p' has a zero in $(-\eta, \eta)$. Since p' is strictly decreasing there, the zero is unique. At the zero, $p'' < 0$, so the stationary point is a strict local maximum.

Remark 10.1 (Why a finite window is used). The minimum of $|\phi'(x)|$ over $\{|x| \geq \eta\}$ is zero because $|\phi'(x)| \rightarrow 0$ in the far tails. Therefore a purely additive uniform derivative error cannot by itself exclude far-tail zeros globally. A global theorem needs an additional tail condition, a weighted norm, or a probability-mass window. For this reason, the threshold theorem is stated on $[-R, R]$ and then paired with a separate tail assumption when a global sign word is required.

Assumption 10.1 (Tail monotonicity for global mode death). For the same $R > 1$, assume

$$p'(x) > 0 \quad (x < -R), \quad p'(x) < 0 \quad (x > R).$$

Corollary 10.2 (Global mode-death threshold). Under the assumptions of Theorem 10.2 and Assumption 10.1, p' has exactly one zero on $\mathbb{R}$. This zero lies in $(-\eta, \eta)$ and is a strict local maximum. If $p(x) \rightarrow 0$ as $|x| \rightarrow \infty$ and $p \geq 0$, then this point is the unique global mode of p .

Proof. Theorem 10.2 gives exactly one zero in $[-R, R]$ and no off-center zeros inside that window. Assumption 10.1 rules out zeros outside $[-R, R]$. Hence there is exactly one stationary point on $\mathbb{R}$, and it is a strict local maximum. Since p' is positive on the far left and has no zero before the unique stationary point, continuity implies $p' > 0$ to the left of that point. Similarly, $p' < 0$ to the right of it. If $p \geq 0$ and $p(x) \rightarrow 0$ at both tails, this increasing-left/decreasing-right structure implies that the unique local maximum is the global maximum.

10.3 Finite- N threshold from derivative local-CLT or Edgeworth error bounds

Let f_N be the standardized convolution density. Suppose a derivative local-CLT or Edgeworth estimate gives constants $A_1, B_1, A_2, B_2 \geq 0$ and the thresholds $\tau_{\eta,R}, \beta_\eta$ from (26)–(27) are used, such that

$$\sup_{|x| \leq R} |f'_N(x) - \phi'(x)| \leq A_1 N^{-1/2} + B_1 N^{-1}, \quad (30)$$

$$\sup_{|x| \leq \eta} |f''_N(x) - \phi''(x)| \leq A_2 N^{-1/2} + B_2 N^{-1}. \quad (31)$$

Define

$$N_\star(\eta, R) := \max \left\{ \left(\frac{2A_1}{\tau_{\eta,R}} \right)^2, \frac{2B_1}{\tau_{\eta,R}}, \left(\frac{2A_2}{\beta_\eta} \right)^2, \frac{2B_2}{\beta_\eta} \right\}. \quad (32)$$

Corollary 10.3 (Explicit finite- N mode-death threshold). If the integer $N \geq 1$ satisfies $N > N_*(\eta, R)$, then

$$\sup_{|x| \leq R} |f'_N(x) - \phi'(x)| < \tau_{\eta, R}, \quad \sup_{|x| \leq \eta} |f''_N(x) - \phi''(x)| < \beta_\eta.$$

Consequently, by Theorem 10.2, f_N has no off-center stationary points in $\eta \leq |x| \leq R$, and has exactly one central stationary point in $(-\eta, \eta)$. Under the tail monotonicity condition, this point is the unique global mode.

Proof. If

$$N > \left(\frac{2A_1}{\tau_{\eta, R}} \right)^2,$$

then $A_1 N^{-1/2} < \tau_{\eta, R}/2$. If

$$N > \frac{2B_1}{\tau_{\eta, R}},$$

then $B_1 N^{-1} < \tau_{\eta, R}/2$. Combining these two inequalities with (30) gives

$$\sup_{|x| \leq R} |f'_N(x) - \phi'(x)| < \tau_{\eta, R}.$$

The same argument with A_2, B_2, β_η gives

$$\sup_{|x| \leq \eta} |f''_N(x) - \phi''(x)| < \beta_\eta.$$

The conclusion follows from Theorem 10.2, and from Corollary 10.2 if the tail condition is also imposed.

10.4 Interpretation of the threshold results

The preceding results may be summarized as follows.

1. **Zero motion.** A simple Gaussian skeleton zero cannot move faster than the derivative-map error divided by the local slope of the Gaussian skeleton map:

$$|z_N - \zeta| \leq \frac{\delta_N}{a}.$$

Here z_N denotes the perturbed finite- N zero, ζ the corresponding Gaussian skeleton zero, δ_N the supremum norm of local derivative-map error, and a the local lower slope bound. For the mode, the skeleton map is ϕ' and $\zeta = 0$. For inflection points, the skeleton map is ϕ'' and $\zeta = \pm 1$.

2. **Mode death.** Additional modes and valleys require off-center zeros of f'_N . Once

$$\sup_{|x| \leq R} |f'_N - \phi'| < \tau_{\eta, R},$$

no such zeros can exist in $\eta \leq |x| \leq R$. Once the central curvature error also satisfies

$$\sup_{|x| \leq \eta} |f''_N - \phi''| < \beta_\eta,$$

the surviving central stationary point is unique and is a strict local maximum.

3. **Finite- N threshold.** If Edgeworth or derivative local-CLT theory gives an explicit bound of order $N^{-1/2} + N^{-1}$, then the threshold $N_*(\eta, R)$ in (32) is a rigorous sufficient condition for finite-window mode death.

The threshold result therefore states: the central limit theorem is used as the limiting reference, while quantitative derivative convergence is converted into explicit finite-window statements about stationary points and inflection points.

Remark 10.2 (Deterministic scope of the threshold argument). The deterministic core of Theorem 10.2 is an affine local skeleton argument. If a perturbed derivative map is within ε of $a(x - \zeta)$, then no zero can lie outside radius η once $\varepsilon < |a|\eta$, and any zero z satisfies $|z - \zeta| \leq \delta/|a|$. The analytic CLT, Edgeworth, Fourier inversion, and derivative local-CLT estimates supply the external convergence estimates used by this deterministic threshold argument.

11 Zero-location perturbation formulas

The preceding theorem gives qualitative skeleton stability. Edgeworth expansion gives quantitative finite- N locations of the skeleton points.

Let $k \in \{1, 2\}$ and suppose x_0 is a simple zero of $\phi^{(k)}$, meaning

$$\phi^{(k)}(x_0) = 0, \quad \phi^{(k+1)}(x_0) \neq 0.$$

For $k = 1$, $x_0 = 0$. For $k = 2$, $x_0 = \pm 1$.

Assume the derivative Edgeworth expansion, uniformly in a neighborhood of x_0 ,

$$f_N^{(k)}(x) = \phi^{(k)}(x) + \rho A^{(k)}(x) + \rho^2 B^{(k)}(x) + o(\rho^2), \quad \rho := N^{-1/2}.$$

For the simple zero continuing from x_0 , seek a zero of $f_N^{(k)}$ in the form

$$x_N = x_0 + \rho a_1 + \rho^2 a_2 + o(\rho^2).$$

Substituting into the expansion and matching powers of ρ gives

$$a_1 = -\frac{A^{(k)}(x_0)}{\phi^{(k+1)}(x_0)}, \tag{33}$$

$$a_2 = -\frac{\frac{1}{2}\phi^{(k+2)}(x_0)a_1^2 + A^{(k+1)}(x_0)a_1 + B^{(k)}(x_0)}{\phi^{(k+1)}(x_0)}. \tag{34}$$

Equations (33)–(34) are the general finite- N perturbation formulas for stationary and inflection locations.

11.1 First-order shifts of the Gaussian skeleton

At first order only $A(x) = (\lambda_3/6) \text{He}_3(x)\phi(x)$ matters. For the mode, take $k = 1$ and $x_0 = 0$. Since

$$(\text{He}_3 \phi)'(0) = -3\phi(0), \quad \phi''(0) = -\phi(0),$$

Equation (33) gives

$$m_N = -\frac{\lambda_3}{2\sqrt{N}} + O(N^{-1}). \quad (35)$$

For the inflection points, take $k = 2$ and $x_0 = \pm 1$. Since

$$(\text{He}_3 \phi)'' = \text{He}_5 \phi, \quad \text{He}_5(1) = 6, \quad \text{He}_5(-1) = -6,$$

and

$$\phi'''(1) = 2\phi(1), \quad \phi'''(-1) = -2\phi(1),$$

Equation (33) gives

$$\iota_{N,-} = -1 - \frac{\lambda_3}{2\sqrt{N}} + O(N^{-1}), \quad \iota_{N,+} = 1 - \frac{\lambda_3}{2\sqrt{N}} + O(N^{-1}). \quad (36)$$

Thus, to first order, skewness translates the central Gaussian skeleton by the same amount $-\lambda_3/(2\sqrt{N})$. Distortions of the spacing between the two inflection points begin at the next order through λ_4 and λ_3^2 .

12 Critical-point dynamics in $\tau = \log N$

Although N is discrete, the logarithmic parameter

$$\tau := \log N$$

may be used for formal interpolation. Let $u(\tau, x)$ be a smooth interpolation of $f_N(x)$, for example through the Edgeworth approximation with $N = e^\tau$. A k th-order critical trajectory satisfies

$$\partial_x^k u(\tau, x_j^{(k)}(\tau)) = 0.$$

Here j indexes the selected critical branch, and $x_j^{(k)}(\tau)$ denotes the location of that branch at logarithmic convolution time τ . If the zero is simple, then

$$\partial_x^{k+1} u(\tau, x_j^{(k)}(\tau)) \neq 0.$$

Implicit differentiation gives

$$\frac{dx_j^{(k)}}{d\tau} = -\frac{\partial_\tau \partial_x^k u(\tau, x_j^{(k)}(\tau))}{\partial_x^{k+1} u(\tau, x_j^{(k)}(\tau))}. \quad (37)$$

Equation (37) gives the formal velocity of a simple critical point. It describes how stationary points and inflection points move as the convolution level increases. When the denominator approaches zero, a simple-zero description breaks down; this corresponds to a birth, death, or merger of critical points.

Using (19) with $N = e^\tau$ gives

$$u(\tau, x) = \phi(x) + e^{-\tau/2} A(x) + e^{-\tau} B(x) + \dots,$$

where the dots denote higher-order asymptotic terms. Here ∂_x and ∂_τ denote partial differentiation with respect to x and τ , respectively. Thus

$$\partial_\tau \partial_x^k u(\tau, x) = -\frac{1}{2} e^{-\tau/2} A^{(k)}(x) - e^{-\tau} B^{(k)}(x) + \dots$$

Therefore, away from merger points, the leading velocity of a critical point is controlled by the skewness correction A and decays at rate $e^{-\tau/2} = N^{-1/2}$.

13 Collapse metrics

The preceding analysis leads to computable quantities. The following definitions translate derivative-map convergence into numerical summaries.

13.1 Mode and stationary counts

Here $\#$ denotes cardinality of a finite zero set. Define the stationary count

$$C_{1,N}(J) := \#\{x \in J : f'_N(x) = 0\}$$

and the mode count

$$M_N(J) := \#\{x \in J : f'_N(x) = 0, f''_N(x) < 0\}.$$

For the Gaussian on $\mathbb{R}$,

$$C_{1,\infty} = 1, \quad M_\infty = 1.$$

A multimodal initial density has $M_N > 1$ for small N in typical examples. Convergence toward the Gaussian skeleton corresponds to the disappearance of additional local maxima and intervening local minima.

13.2 Inflection count

Define

$$C_{2,N}(J) := \#\{x \in J : f''_N(x) = 0\}.$$

For the Gaussian on $\mathbb{R}$,

$$C_{2,\infty} = 2.$$

The inflection count is informative because a distribution may become unimodal before its curvature structure has fully stabilized.

13.3 Gaussian skeleton distance

When the main stationary point m_N and main inflection points $\iota_{N,-}, \iota_{N,+}$ are identified, define

$$D_N := |m_N| + |\iota_{N,-} + 1| + |\iota_{N,+} - 1|. \quad (38)$$

A refined version may use the first-order skewness-corrected skeleton:

$$D_N^{\text{skew}} := \left| m_N + \frac{\lambda_3}{2\sqrt{N}} \right| + \left| \iota_{N,-} + 1 + \frac{\lambda_3}{2\sqrt{N}} \right| + \left| \iota_{N,+} - 1 + \frac{\lambda_3}{2\sqrt{N}} \right|. \quad (39)$$

This quantity separates finite- N skewness translation from deeper shape residues.

13.4 Gaussian map completion time

For a window J on which the exact sign word $W_J(f_N)$ is defined, an analytic completion index can be written as

$$N_{\text{map}}(J) := \min \{N_0 \in \mathbb{N} : W_J(f_n) = I \rightarrow IV \rightarrow III \rightarrow II \text{ for every } n \geq N_0\}, \quad (40)$$

Here $\mathbb{N}$ denotes the positive integers. The definition is used only when the set on the right-hand side is nonempty. Under the derivative local-CLT assumptions and on a fixed central window with the transition neighborhoods excluded as in Theorem 9.1, such eventual completion follows for sufficiently large N .

In computation, the exact word is replaced by a tolerance-dependent word obtained from a finite grid and root-refinement rule. The resulting numerical completion index is denoted by $N_{\text{map}}(J, \varepsilon)$, where ε records the zero-detection and sign-stability tolerance. This numerical quantity should be interpreted as an approximation to (40), not as a separate mathematical theorem.

Remark 13.1 (Mode count need not be monotone). The count M_N or $C_{1,N}$ should not be assumed to decrease monotonically for every possible initial distribution and every finite N . The derivative-map representation tracks the realized finite- N path rather than imposing monotonicity. The robust claim is eventual central-window stabilization under regularity assumptions, not monotone collapse at every step.

14 Discrete distributions and smoothing

If X is discrete, then Z_N is generally lattice-valued and does not have an ordinary density. Derivatives such as f'_N and f''_N are not directly defined. There are two standard ways to extend the construction.

14.1 Gaussian mollification

Let p_N denote the law of Z_N as a discrete probability measure. For bandwidth $h > 0$, define

$$\phi_h(x) := \frac{1}{h} \phi\left(\frac{x}{h}\right), \quad f_{N,h} := p_N * \phi_h.$$

Here $*$ denotes convolution. Then $f_{N,h}$ is smooth, and its characteristic function is

$$\varphi_{N,h}(t) = \varphi_{Z_N}(t) e^{-h^2 t^2 / 2}. \quad (41)$$

The derivative-map construction applies to $f_{N,h}$:

$$\mathcal{S}_{N,h} := \{x : f'_{N,h}(x) = 0\}, \quad \mathcal{I}_{N,h} := \{x : f''_{N,h}(x) = 0\}.$$

If h is fixed, the limiting smoothed density is $N(0, 1 + h^2)$. Its inflection points are at $\pm\sqrt{1 + h^2}$. If one wants the unsmoothed $N(0, 1)$ endpoint, take $h = h_N \rightarrow 0$ while keeping h_N larger than the lattice spacing scale. For a unit-lattice original variable, the standardized lattice spacing is of order $N^{-1/2}$, so a practical asymptotic regime is

$$N^{-1/2} \ll h_N \ll 1.$$

14.2 Finite-difference sign maps

Alternatively, the probability mass function on its lattice can be analyzed directly by replacing derivatives with finite differences. This produces a discrete stationary–inflection map. The continuous derivative map is usually more direct for comparison with the Gaussian endpoint, while the finite-difference version is useful for purely discrete applications.

15 Numerical implementation

The numerical procedure can be implemented without repeated grid-based convolution. The companion script implements the diagnostic pipeline using `numpy`, `pandas`, and `matplotlib`. The implementation repository is available at https://github.com/DavidHShen/Derivative-Map_Dynamics_CLT. For the built-in Gaussian-mixture examples, the default route uses exact finite Gaussian-mixture convolution, which is faster and avoids Fourier-truncation error. A direct Fourier-quadrature route remains available as an optional diagnostic mode so that the connection between characteristic functions and derivative reconstruction remains visible.

1. **Input.** Specify a finite Gaussian-mixture law through its component weights, means, and variances. The surrounding formulation also explains how the same derivative-map diagnostics can be applied to other smooth densities, characteristic functions, smoothed discrete laws, or smoothed empirical laws when the required density derivatives can be reconstructed reliably.
2. **Standardize.** Compute μ and σ , and form the characteristic function (10) for Z_N .
3. **Density and derivative reconstruction.** For the built-in finite Gaussian mixtures, compute f_N , f'_N , and f''_N by the exact multinomial Gaussian-mixture expansion of the N -fold convolution. In optional Fourier mode, choose an x -grid and a compatible finite frequency interval $[-t_{\max}, t_{\max}]$, and approximate

$$f_N^{(k)}(x) \approx \frac{1}{2\pi} \int_{-t_{\max}}^{t_{\max}} (-it)^k e^{-itx} \varphi_{Z_N}(t) dt, \quad k = 0, 1, 2,$$

by trapezoidal quadrature.

4. **Zero detection.** Locate zeros of f'_N and f''_N by bracketed sign changes on a significant-density window with adaptive derivative tolerances. Classify stationary points by the sign of f''_N at the detected zeros.
5. **Sign word.** Sort the union of the detected zeros and record the quadrant label $Q(\text{sgn } f'_N, \text{sgn } f''_N)$ on each interval.
6. **Metrics.** Compute stationary counts, mode counts, valley counts, inflection counts, sign words, L^2 distances to the Gaussian density and its first two derivatives, and a simple Gaussian-skeleton distance comparing detected roots with the target skeleton $\{0\}$ and $\{-1, 1\}$.

Numerically, tail regions must be treated with a density threshold. For a chosen tail-mass tolerance $\alpha \in (0, 1)$, a natural window is the smallest interval $J_N(\alpha)$ satisfying

$$\int_{J_N(\alpha)} f_N(x) dx \geq 1 - \alpha,$$

for a small value such as $\alpha = 10^{-6}$ or 10^{-8} . In the implementation, zero detection is performed only on grid points where the reconstructed density exceeds a combined absolute and relative floor. This avoids spurious sign flips in regions where f_N , f'_N , and f''_N are below numerical precision. All reported counts and sign words should therefore be read as window- and tolerance-dependent diagnostics, not as global topological claims.

Scientific-computing safeguards. The implementation follows a conservative numerical-audit protocol. First, the standard Gaussian characteristic function is reconstructed as a self-test, and the detected skeleton must contain one stationary point near 0 and two inflection points near ± 1 . Second, every reported run records mass, mean, variance, negative-density mass, endpoint density and derivative magnitudes, and, in Fourier mode, high-frequency edge indicators for the multipliers $|t|^k \varphi_{Z_N}(t)$, $k = 0, 1, 2$. Third, root counts use bracketed sign changes on the significant-density window rather than accepting every near-zero tail grid value as a zero. Fourth, the exact-mixture and Fourier routes are separated so that the examples have a stable benchmark while still preserving a characteristic-function implementation path. These checks do not turn tolerance-dependent counts into exact topological claims; rather, they make the numerical claims reproducible and auditable.

16 Representative test distributions

The mathematical construction can be illustrated on several controlled examples.

16.1 Symmetric two-Gaussian mixture

Let

$$X \sim \frac{1}{2}N(-a, s^2) + \frac{1}{2}N(a, s^2).$$

Here $a > 0$ is the separation parameter and $s > 0$ is the common component standard deviation. Then $\mu = 0$ and $\sigma^2 = a^2 + s^2$. The distribution is bimodal when a is sufficiently large relative to s . The characteristic function is

$$\varphi_X(t) = e^{-s^2 t^2 / 2} \cos(at).$$

The standardized-sum characteristic function is

$$\varphi_{Z_N}(t) = \left[\exp\left(-\frac{s^2 t^2}{2N(a^2 + s^2)}\right) \cos\left(\frac{at}{\sqrt{N(a^2 + s^2)}}\right) \right]^N.$$

This example begins with two separated modes and has an explicit characteristic function. Symmetry implies $\lambda_3 = 0$, so the first-order skeleton translation vanishes.

16.2 Asymmetric mixture

Let

$$X \sim wN(m_1, s_1^2) + (1 - w)N(m_2, s_2^2), \quad 0 < w < 1.$$

Here $m_1, m_2 \in \mathbb{R}$ are component means and $s_1, s_2 > 0$ are component standard deviations. This example introduces skewness and unequal mode weights. It can be used to test the first-order shift formulas (35)–(36) and separates skewness translation from true shape residue.

16.3 Discrete multi-mass distribution

Let X take finitely many values $x_1, \dots, x_L$ with probabilities $p_1, \dots, p_L$, where $p_\ell > 0$ and $\sum_{\ell=1}^L p_\ell = 1$. Its characteristic function is

$$\varphi_X(t) = \sum_{\ell=1}^L p_\ell e^{itx_\ell}.$$

The smoothed derivative map uses (41). The example illustrates how discrete multi-mass structures become smooth Gaussian-like densities after convolution and mollification.

17 Companion numerical outputs

The companion implementation was run on three representative examples: a symmetric two-Gaussian mixture, an asymmetric three-Gaussian mixture, and a smoothed discrete multi-mass law. The results are meant to illustrate the diagnostic objects defined in the paper rather than to establish a universal monotonic law in N . In particular, mode and inflection counts can temporarily increase under convolution before collapsing on the chosen central window, especially for sharply separated or lightly smoothed input laws.

Figure 4 shows the density-level collapse for the three examples. Figure 5 shows representative derivative maps at $N = 1$ and $N = 60$ for the symmetric two-Gaussian mixture. The later map is close to the Gaussian stationary-inflection skeleton on the selected window, while the early map contains additional derivative zeros. Figure 6 reports the simple Gaussian-skeleton diagnostic distance used by the script.

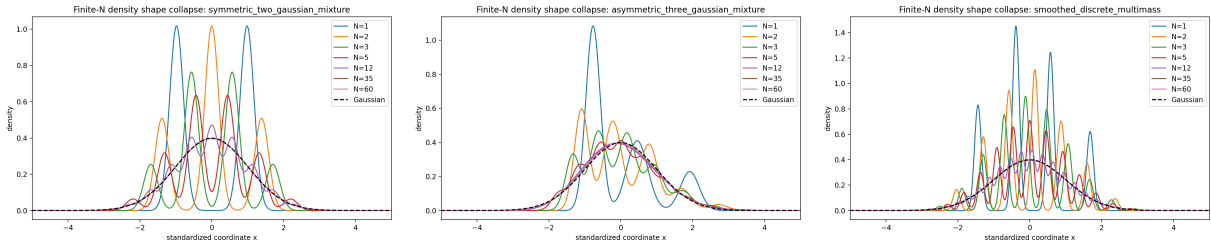


Figure 4: Finite- N density-shape collapse for the three companion examples: symmetric two-Gaussian mixture, asymmetric three-Gaussian mixture, and smoothed discrete multi-mass law. The Gaussian density is included as the dashed reference curve. The digital version may be enlarged for detailed inspection.

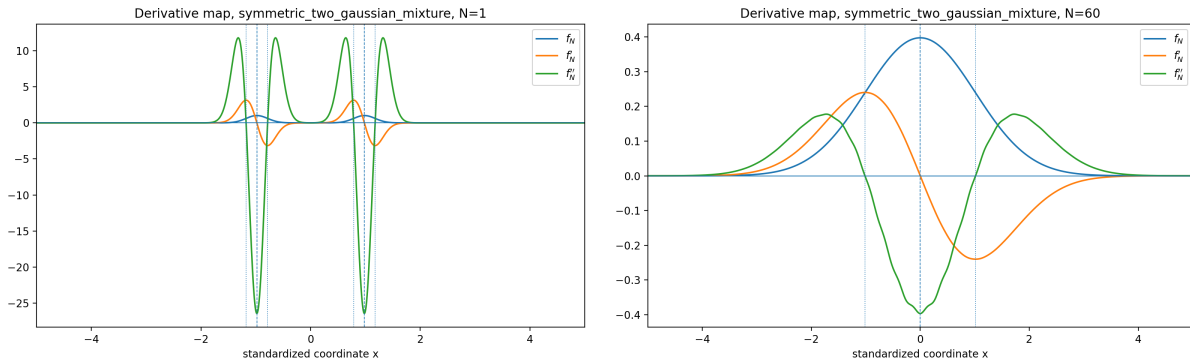


Figure 5: Derivative maps for the symmetric two-Gaussian mixture at $N = 1$ and $N = 60$. Solid curves show f_N , f'_N , and f''_N ; vertical markers indicate detected stationary and inflection zeros. The digital version may be enlarged for detailed inspection.

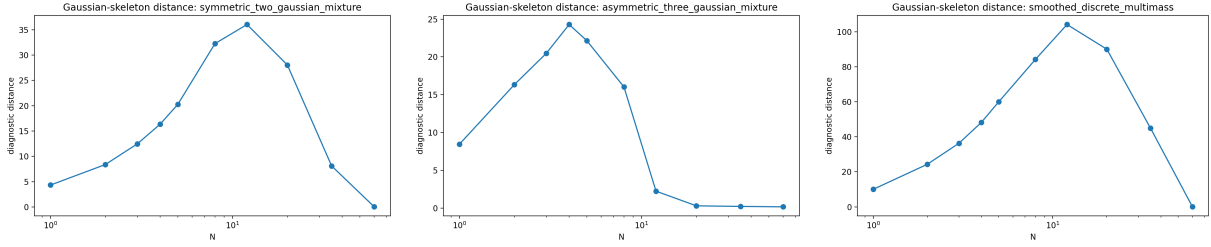


Figure 6: Gaussian-skeleton diagnostic distance for the three examples. The metric is diagnostic rather than canonical: it measures the detected root configuration against the Gaussian target roots 0 and ± 1 , with penalties for extra roots on the selected central window. The digital version may be enlarged for detailed inspection.

Table 1 records selected outputs from the generated CSV tables. The examples show two important features of the proposed representation. First, density-level L^2 distance to the Gaussian decreases substantially as N grows. Second, derivative-level topology may be less monotone at intermediate N , so the mode-death and skeleton-stabilization diagnostics should be evaluated on the chosen finite window rather than inferred only from distributional convergence.

Example	N	Modes	Inflections	Skeleton distance	$L^2(f_N, \phi)$
Symmetric two-Gaussian mixture	1	2	4	4.334	0.7131
Symmetric two-Gaussian mixture	12	7	26	36.023	0.0769
Symmetric two-Gaussian mixture	60	1	2	0.030	0.0018
Asymmetric three-Gaussian mixture	1	3	6	8.472	0.5129
Asymmetric three-Gaussian mixture	12	1	4	2.244	0.0325
Asymmetric three-Gaussian mixture	60	1	2	0.171	0.0139
Smoothed discrete multi-mass law	1	4	8	10.064	0.7369
Smoothed discrete multi-mass law	12	27	54	104.142	0.0779
Smoothed discrete multi-mass law	60	1	2	0.037	0.0024

Table 1: Selected companion implementation metrics on the central analysis window. Counts and distances are tolerance-dependent numerical diagnostics generated under the density-thresholded zero-detection rule; they are not asserted as exact global mode counts.

18 Applications

The approach is a mathematical and numerical representation, not a new probability limit theorem. It is relevant in settings where repeated aggregation, averaging, convolution, smoothing, or Gaussian approximation is used, while finite- N density shape remains important.

18.1 Statistical learning and distribution diagnostics

In statistical learning systems, features, residuals, noise components, mini-batch averages, ensemble outputs, and aggregated latent variables are often approximated by Gaussian laws. The derivative-

map representation tests whether the approximation is only distributional or whether the density shape has entered the Gaussian stationary–inflection skeleton on a specified window. The output identifies concrete geometric deviations: additional modes, additional valleys, unstable inflection points, or incorrect quadrant words.

18.2 Density topology, KDE, and generative-model evaluation

The zero sets of f' and f'' are topological summaries of one-dimensional density shape. In kernel density estimation and generative modeling, a model may match moments or likelihood while creating additional modes or losing important modes. The stationary–inflection skeleton can therefore be used as a shape-based criterion for over-smoothing, mode loss, and non-Gaussian finite-sample residue.

18.3 Monte Carlo and numerical probability

Many simulations use Gaussian approximations for sums. The threshold $N_\star$ gives a finite-sample criterion for when a Gaussian approximation is not only distributionally close but also stable at the derivative-map level. The skewness-corrected distance (39) can distinguish ordinary finite- N skewness from more serious multimodal residue.

18.4 Signal processing and aggregation systems

Repeated averaging and filtering often produce Gaussian-like outputs. The derivative-map representation provides one way to quantify how multi-peak input noise becomes a one-peak output distribution. This may be relevant in systems where transient multimodality indicates regime switching, sensor mixture, or unresolved latent classes.

18.5 Financial risk aggregation

Portfolio returns, risk-factor changes, and P&L components are often aggregated across instruments, horizons, or scenarios. When the components are mixture-like or regime-dependent, finite- N aggregation may still contain multiple modes even if the limiting intuition suggests Gaussian behavior. The stationary–inflection metrics provide one way to assess whether risk aggregation has reached a Gaussian-like density shape or still contains finite- N regime structure.

19 Scope and limitations

Several limitations are inherent in the formulation.

1. The construction is a finite- N derivative-map representation, not a replacement for the central limit theorem.
2. Global eventual unimodality is not asserted for arbitrary distributions without tail and regularity assumptions; this distinction is consistent with the separate theory of unimodality and strong unimodality under convolution [4]. The rigorous core is central-window skeleton stability under derivative local-CLT conditions.

3. Mode counts and sign words may not change monotonically in N . The method tracks finite- N paths rather than assuming monotonicity.
4. Numerical implementation requires careful grid design, Fourier truncation, smoothing choices for discrete laws, and density thresholds in far tails.
5. In higher dimensions, the analogue of stationary and inflection points becomes the geometry of gradients, Hessians, ridges, and level sets. The one-dimensional case is treated first because it admits an explicit derivative sign word.

20 Conclusion

The finite- N central-limit process can be analyzed not only as convergence of distributions, but also as convergence of derivative-level density geometry. A multimodal initial law may produce standardized convolution densities with additional stationary points, valleys, and inflection points before the Gaussian profile is approached. Under the stated smoothness, moment, Fourier-inversion, finite-window, and tail conditions, higher cumulants decay on the characteristic-function side, and derivative local-CLT or Edgeworth estimates transfer this decay to the density side. The derivative map (f'_N, f''_N) then records the associated geometric stabilization: off-center stationary points are excluded once the derivative error is below explicit thresholds, simple skeleton zeros move within controlled neighborhoods, and the central-window sign word becomes the Gaussian stationary–inflection word

$$\mathcal{S}_\infty = \{0\}, \quad \mathcal{I}_\infty = \{-1, 1\}, \quad W_\infty = I \rightarrow IV \rightarrow III \rightarrow II.$$

Here $\mathcal{S}_\infty$ denotes the limiting Gaussian stationary set, $\mathcal{I}_\infty$ denotes the limiting Gaussian inflection set, and W_∞ denotes the limiting Gaussian derivative sign word. The construction is therefore best viewed as a characteristic-function-driven derivative-map representation of finite- N central-limit shape convergence. Its main role is diagnostic and structural: it provides sufficient finite-window criteria indicating when an aggregated density has entered the Gaussian stationary–inflection skeleton, and when residual finite- N multimodal features remain visible on the chosen analysis window. The companion implementation supports this interpretation by converting the definitions into reproducible density, derivative-map, zero-count, sign-word, and skeleton-distance outputs for controlled examples.

The threshold analysis in Section 10 supplies the main deterministic step behind this conclusion. It shows that derivative convergence becomes geometrically effective only after it is translated into finite-window inequalities: zero-motion estimates around the Gaussian skeleton and exclusion bounds for off-center stationary points. These estimates connect asymptotic Gaussian convergence to a checkable finite- N mode-death criterion, which is the central technical mechanism supporting the paper’s shape-collapse interpretation.

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Appendix

A Hermite identities used in the derivations

The probabilists' Hermite polynomials satisfy

$$\phi^{(m)}(x) = (-1)^m \text{He}_m(x)\phi(x).$$

Therefore

$$\text{He}_m(x)\phi(x) = (-1)^m \phi^{(m)}(x).$$

For $m = 3$,

$$\text{He}_3(x)\phi(x) = -\phi'''(x).$$

Consequently

$$(\text{He}_3 \phi)' = -\phi^{(4)}, \quad (\text{He}_3 \phi)'' = -\phi^{(5)} = \text{He}_5 \phi.$$

These identities are used in the zero-shift calculations.

B Derivation of the first-order zero shifts

Let $\rho = N^{-1/2}$ and

$$f_N^{(k)}(x) = \phi^{(k)}(x) + \rho A^{(k)}(x) + O(\rho^2).$$

Suppose $x_N = x_0 + \rho a_1 + O(\rho^2)$ and $\phi^{(k)}(x_0) = 0$. Taylor expansion gives

$$0 = f_N^{(k)}(x_N) = \rho \left[\phi^{(k+1)}(x_0)a_1 + A^{(k)}(x_0) \right] + O(\rho^2).$$

Hence

$$a_1 = -\frac{A^{(k)}(x_0)}{\phi^{(k+1)}(x_0)}.$$

For the mode, $k = 1$, $x_0 = 0$, and

$$A'(0) = \frac{\lambda_3}{6} (\text{He}_3 \phi)'(0) = \frac{\lambda_3}{6} [-3\phi(0)] = -\frac{\lambda_3}{2} \phi(0).$$

Since $\phi''(0) = -\phi(0)$,

$$a_1 = -\frac{A'(0)}{\phi''(0)} = -\frac{\lambda_3}{2}.$$

Therefore

$$m_N = -\frac{\lambda_3}{2\sqrt{N}} + O(N^{-1}).$$

For the right inflection point, $k = 2$, $x_0 = 1$, and

$$A''(1) = \frac{\lambda_3}{6} (\text{He}_3 \phi)''(1) = \frac{\lambda_3}{6} \text{He}_5(1) \phi(1) = \lambda_3 \phi(1).$$

Since $\phi'''(1) = 2\phi(1)$,

$$a_1 = -\frac{A''(1)}{\phi'''(1)} = -\frac{\lambda_3}{2}.$$

The left inflection point gives the same first-order shift. Therefore both inflection points are translated by $-\lambda_3/(2\sqrt{N})$ at leading order.